

Project Design Phase-II Data Flow Diagram & User Stories

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Team ID	SWTID-2026-7735
Project Name	AI Blog Nest API
Maximum Marks	4 Marks

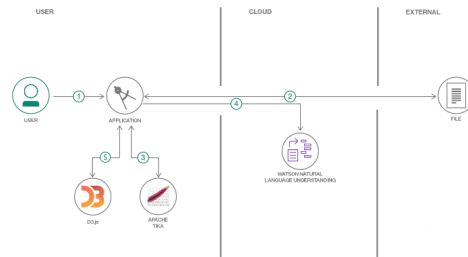
Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

AI Blog Nest API – Level 0 Data Flow Diagram

Example: DFD Level 0 (Industry Standard)

Flow



1. User configures credentials for the Watson Natural Language Understanding service and starts the app.
2. User selects data file to process and load.
3. Apache Tika extracts text from the data file.
4. Extracted text is passed to Watson NLU for enrichment.
5. Enriched data is visualized in the UI using the D3.js library.

User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Reader	Registration & Authentication	USN-1	As a reader, I can register with my email and password so that I can access the application.	Valid account is created and credentials are accepted.	High	Sprint-1
Reader	Login	USN-2	As a reader, I can log in with valid credentials and receive an authenticated token.	Valid token is issued; invalid credentials are rejected.	High	Sprint-1
Reader	Blog Discovery	USN-3	As a reader, I can view published blogs.	Only accessible/published content is returned.	High	Sprint-1
Reader	Categorization & Search	USN-4	As a reader, I can filter blogs by category or tag.	Matching posts are returned for valid filters.	Medium	Sprint-1
Reader	Comments	USN-5	As a reader, I can submit a comment on a blog.	Comment is stored with user/blog references and moderation state.	Medium	Sprint-2

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Author	Blog CRUD	USN-6	As an author, I can create a blog with content, status, tags and category.	Valid blog is stored with required metadata.	High	Sprint-1
Author	Blog CRUD	USN-7	As an author, I can update my blog content and metadata.	Authorized owner can update permitted fields.	High	Sprint-1
Author	Blog CRUD	USN-8	As an author, I can delete my blog when it is no longer needed.	Authorized delete request is processed successfully.	High	Sprint-2
Author	AI Blog Generation	USN-9	As an author, I can provide a theme and generate a structured blog using AI.	LLM returns content that can be reviewed before saving.	High	Sprint-2
Author	AI Summarization	USN-10	As an author, I can generate a concise summary from a long article.	System returns an abstract/metadata summary.	High	Sprint-2
Author	Categorization	USN-11	As an author, I can assign categories and tags to a blog.	Valid category/tag references are associated with the blog.	Medium	Sprint-2
Author	Asset Assignment	USN-12	As an author, I can associate supported media assets with a blog.	Asset reference is validated and linked.	Medium	Sprint-2
Administrator	Comment Moderation	USN-13	As an administrator, I can mark comments approved, pending or spam.	Authorized moderator can persist the new moderation state.	High	Sprint-2
Administrator	Role Management	USN-14	As an administrator, I can access protected management routes according to my role.	Unauthorized roles receive an access error.	High	Sprint-1
Administrator	User Management	USN-15	As an administrator, I can manage user access according to role permissions.	Only authorized admin actions are accepted.	Medium	Sprint-3
Administrator	Content Management	USN-16	As an administrator, I can manage permitted blog status and settings.	Authorized changes are persisted.	High	Sprint-2
System	Input Security	USN-17	As the system, I must sanitize and validate payloads before processing.	Invalid/malicious payloads are rejected or sanitized.	High	Sprint-1
System	NoSQL Injection Protection	USN-18	As the system, I must prevent unsafe query parameters from manipulating database operations.	Unsafe operators/parameters are blocked or sanitized.	High	Sprint-1
System	AI Failure Handling	USN-19	As the system, I must handle LLM failures without corrupting blog data.	Controlled error is returned and invalid content is not saved.	Medium	Sprint-2
System	API Performance	USN-20	As the system, I should use indexed queries and pagination for	Large queries return bounded result	Medium	Sprint-3

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
			large collections.	sets efficiently.		

College: C.T.T.E College for Women

Team Leader: Natshitha.P

Team Members: Nivedha.A, Sanmathi.B, Merlin.Y

DFD Data Stores & Interfaces: D1 MongoDB stores users, roles, blogs, comments, categories/tags and metadata. D2 Asset Storage stores media. D3 External LLM receives approved prompts/context and returns generated or summarized content. Authentication and role checks occur before protected operations.